

Week 5: Kinematics Problem-Solving Session

Instructor Notes and Worked Solutions

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1 Purpose

This Week 5 document replaces the usual theory notes with a guided problem-solving session covering the core manipulator-kinematics material from Weeks 3 and 4:

- forward kinematics with DH tables
- geometric forward kinematics for articulated arms
- ZYZ Euler angle extraction (previews the wrist sub-problem)
- analytical inverse kinematics for planar, SCARA, and elbow robots
- kinematic decoupling for a spherical wrist

The original numerical inverse-kinematics theory files were preserved in:

`weeks/W05_numerical-inverse-kinematics/theory/archive/numerical-ik-original/`

2 Session Structure

2.1 Recommended Timing

	Problem	Topic	Time
1	2R planar FK with DH		15 min
2	SCARA FK		20 min
3	Spherical wrist FK + ZYZ extraction		25 min
4	3R elbow-arm FK position		20 min
5	2R planar IK		20 min
6	SCARA IK (inverse of Problem 2)		25 min
7	3R elbow-arm IK		25 min
8	6-DOF wrist decoupling		30 min

2.2 FK/IK Pairings

Three FK/IK problem pairs reinforce the forward-inverse connection:

- Problems 2 and 6: SCARA FK $\rightarrow$ SCARA IK (same robot, same target)
- Problems 4 and 7: 3R elbow FK $\rightarrow$ 3R elbow IK (same robot, same target)
- Problems 3 and 8: ZYZ forward $\rightarrow$ ZYZ extraction (same formulas)

Explicitly point out these connections when transitioning between Part A and Part B.